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Review article

Comparison of digital technologies for occlusal analysis in dentate arches: A systematic review

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ABSTRACT

Objective: This systematic review evaluates the accuracy, reproducibility, and clinical applications of digital occlusal analysis systems in dentate arches.

Methods: MEDLINE/PubMed, Scopus, Web of Science, and Embase databases were searched for articles published up to September 2024. Inclusion criteria encompassed in vitro and clinical studies comparing digital occlusal analysis systems, including T-Scan, OccluSense, Accura, and intraoral scanners.

Results: A total of 3354 articles were identified, of which eight were included. Studies consistently demonstrated superior accuracy and reproducibility of T-Scan systems in dynamic occlusal force assessment. Cerec Omnicam provided reliable static occlusal evaluations, particularly in anterior regions, while OccluSense was an affordable alternative, although reproducibility across various operators and settings was limited. The integration of intraoral scanners with digital occlusal analyzers significantly enhanced diagnostic precision, allowing the simultaneous visualization of occlusal contacts, force distribution, and dynamic sequences. However, variability in the calibration methods and reporting metrics was noted across studies.

Conclusions: Digital occlusal analysis systems, particularly T-Scan, deliver enhanced accuracy and reproducibility for occlusal assessment. Combining systems such as T-Scan with intraoral scanners (Trios or Medit I600) optimizes the clinical outcomes by integrating dynamic force analyses with precise static visualization. Future research should prioritize standardizing methodologies and conducting robust clinical trials to establish comprehensive clinical guidelines.

Clinical significance: This systematic review provides clinicians with evidence-based guidance to select appropriate digital occlusal analysis technologies, which can enhance diagnostic accuracy and patient care.

1. Introduction

Occlusion plays a critical role in dental health and is a key factor in both diagnosis and prognosis. The Glossary of Prosthodontic Terms defines occlusion as the static relationship between the incising or masticating surfaces of the maxillary or mandibular teeth or tooth

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analogs [1]. Occlusal contacts and masticatory forces provide essential information about a patient's chewing efficiency and overall function [2,3]. Studies have reported a strong correlation between the number and area of occlusal contacts and chewing efficiency, underscoring the importance of accurately localizing occlusal contacts in the clinical environment [4]. Precise analysis of the occlusal contacts is also essential for identifying interferences during occlusal adjustments, which, if left uncorrected, can lead to significant functional disturbances. These disturbances may include changes in the tooth-supporting tissues, masticatory muscles, and the temporomandibular joint (TMJ) [5,6]. Therefore, an accurate and thorough analysis of occlusal relationships is essential for maintaining long-term stability and functionality in both natural dentition and prosthetic restorations.

Occlusal contact point~~s~~ analysis using conventional methods is a well-established approach in dentistry [7]. These methods include tools, such as articulating paper, silk strips, occlusal sprays, transillumination, early contact indicators, and occlusal sonography [8,9]. Among these, articulating paper is the most widely used tool for visualizing and evaluating occlusal contacts. However, these methods are limited to static occlusion and fail to account for dynamic mandibular movements, such as maximum intercuspation, protrusion, and laterotrusion [10]. Both qualitative and quantitative methods are employed to register and evaluate occlusal contacts [11]. Conventional methods allow for visual determination of the occlusal contact size, number, and intensity, which are qualitative parameters. However, they may not provide precise quantitative data, leading to variability in results depending on the material used. Most conventional methods are successful in determining the location of tooth contacts but fail to reliably measure the forces of these contacts [12]. Additionally, these methods are largely based on subjective evaluations, increasing the risk of inconsistencies and operator-dependent errors in clinical practice [13].

Advancements in digital dentistry have improved the accuracy and reliability of occlusal contact registration [14]. Digital occlusal indicators vary in design and function and are based on different operating principles. Systems such as the Dental Prescale II (DP2, GC Corp.) quantitatively assess the bite force for the entire dentition, consequently evaluating masticatory function [15]. However, they are limited by their inability to capture the sequential order of occlusal contacts [16]. The T-Scan system, in contrast, offers dynamic occlusal analysis by recording the timing and intensity of occlusal contacts in real-time from initial contact to maximum intercuspation [17,18].

More recent systems, such as Accura and OccluSense, have been introduced as cost-effective alternatives to T-Scan; they integrate articulating paper with digital recording to provide additional occlusal contact information [18,19]. However, few studies have assessed their precision and reproducibility [20]. Additionally, computer-aided design (CAD)/ computer-aided manufacturing (CAM) systems and intraoral scanners (IOS) have been integrated into occlusal analysis workflows, allowing practitioners to digitally evaluate and refine occlusion during the design phase of dental restorations [8,14]. These technologies facilitate three-dimensional modeling of dental arches and occlusal surfaces, enabling a more precise assessment of occlusion, particularly in prosthetic and orthodontic applications [9,21,22].

Despite these advancements, comprehensive comparative studies evaluating the accuracy, reliability, and clinical efficacy of various digital occlusal assessment technologies remain scarce limited. Thus, this systematic review aims to evaluate the effectiveness of different digital occlusal methods employed in dentate arches.

2. Material and methods

This systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-analyses checklist (PRISMA; https://www.prisma-statement.org/) and has been registered in the PROSPERO international prospective register of systematic reviews (CRD42023488428; https://www.crd.york.ac.uk/prospero/). To guide this review, the following research question was formulated: "Which digital occlusal assessment technologies provide the most accuracy and reproducibility results for occlusal analysis in dentate arches?" The PICOS framework was applied as follows: participants (P) were individuals with dentate arches, indicating the presence of natural teeth. Intervention (I) was the use of various digital occlusal assessment technologies (e.g., T-Scan, Dental Prescale II, Accura, OccluSense, IOS, and CAD/CAM systems); Comparison (C) was established among different digital occlusal assessment methods, excluding traditional occlusal evaluation techniques. The primary outcome measures (O) included accuracy, reliability, and clinical efficacy of occlusal assessments, along with the advantages and disadvantages of these technologies in clinical applications. The study design (S) considered for inclusion comprised in vitro studies and clinical studies (Table 1).

2.1. Search strategy

The PubMed, Web of Science, Scopus, and Embase databases were searched up to September 2024. Inclusion criteria comprised in vitro studies and clinical studies comparing different digital occlusal assessment methods, written in English. Exclusion criteria included reviews, animal studies, case reports, and technique reports. After removing duplicates~~,~~ two authors (G.D., C.N.C.W.) independently screened the titles and abstracts using a pre-established screening form based on the inclusion and exclusion criteria (Table 2). The screening process was conducted using Rayyan software (Qatar Computing Research Institute, Doha, Qatar) [23]. Differences in article selection were resolved through discussion between the two reviewers. If consensus was not reached, a third author (I.N.R.R.) independently assessed the study and provided the final decision.. Additionally, as part of the search methodology, the reference lists of all the included studies were manually reviewed (i.e., a snowballing strategy) to identify any additional potentially relevant articles not retrieved through the initial database search. Search queries, including Boolean operators for each database, are provided in Supplementary Table S1 to ensure transparency and reproducibility. No restrictions were applied regarding the year of publication.

2.2. Eligibility assessment and calibration

Before the formal screening, a calibration exercise was performed between the two reviewers (G.D. and C.N.C.W.) using a random sample of 100 records to ensure consistent interpretation of the eligibility criteria. Discrepancies were discussed and resolved by consensus. This process was followed by independent screening of all titles, abstracts, and full texts. Any disagreement during the full review process was resolved through discussion or consultation with a third reviewer (I.N.R.R.).

2.3. Data extraction

The same two authors (G.D., C.N.C.W.) independently extracted the data from the identified studies. A standardized extraction form was pilot-tested on a subset of studies to ensure consistency between

Table 1
PICOS criteria used for study inclusion.

Component	Criteria
P (Participants)	Individuals with dentate arches (presence of natural teeth)
I (Intervention)	Use of various digital occlusal assessment methods
C (Comparison)	Comparison between different digital occlusal assessment methods
O (Outcome measures)	Accuracy, reliability, and clinical efficacy of occlusal assessments; advantages and disadvantages of digital methods in clinical practice
S (Study design)	In vitro studies, clinical studies (randomized controlled trials, prospective, retrospective, observational)

Table 2
Exclusion criteria.

Category	Exclusion Criteria
Participants	Edentulous patients, partially edentulous patients, presence of dental implants, or wear of fixed/removable dentures
Intervention	Use of conventional (non-digital) occlusal assessment methods; incomplete or unclear documentation of the digital assessment protocol; conditions preventing accurate assessment (e.g., severe malocclusion)
Comparison	Studies evaluating only a single digital method without comparison; comparisons between digital and conventional methods only; studies focused solely on manual/analogue assessment techniques
Outcome measures	Studies not reporting on accuracy, reliability, or clinical effectiveness; studies evaluating outcomes unrelated to occlusion (e.g., esthetics)
Study design	Reviews, letters, abstracts, posters, protocols, personal opinions, technique articles, case reports, case series, animal studies, non-English articles, or studies without full text access

reviewers prior to complete data extraction. The following information was collected when available: authors, year of publication, study design, number of patients and phantoms, and details of the digital technology used, including IOS, desktop scanners (DS), and digital occlusion analysis systems (DOAS). Additional extracted data included sensor thickness, device calibration, scan parameters, bite positions (maximum intercuspal position [MIP], centric relation position [CRP]), calibration of models or patients, superimposition software and methodology, anatomical references, measurement procedures, and reliability measures such as the intraclass and inter-examiner correlation coefficients (IECC). Relevant outcomes, key findings, and the main conclusions of each study were also recorded. In cases of missing essential data, the study authors were contacted for additional information.

2.4. Risk of bias assessment

The risk of bias in the selected studies was assessed using tools appropriate to their study designs. For cross-sectional studies included in this review, the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Prevalence Studies was used to assess the risk of bias. This tool evaluates nine domains, including the appropriateness of the sample frame, sampling method, sample size, description of the study participants and settings, data analysis coverage, validity of methods for identifying the condition, reliability of measurements, statistical analysis, and adequacy of response rates. Each domain was rated as "Yes," "No," or "Unclear (U)," and the total score was calculated as the percentage of "Yes" responses. The risk of bias was categorized as low (≥70 %), moderate (50–69 %), or high (≤49 %), providing a structured assessment of potential biases and methodological quality [24]. The QUIN Tool (Quality Index for In Vitro Studies) was used to assess the methodological quality and potential bias of the included in vitro studies. This tool evaluates domains such as clarity of objectives, sample size justification, sampling techniques, comparison group details, methodological transparency, operator details, randomization, outcome measurement methods, blinding, statistical analysis, and presentation of the results. Scores are converted into percentages to classify the risk of

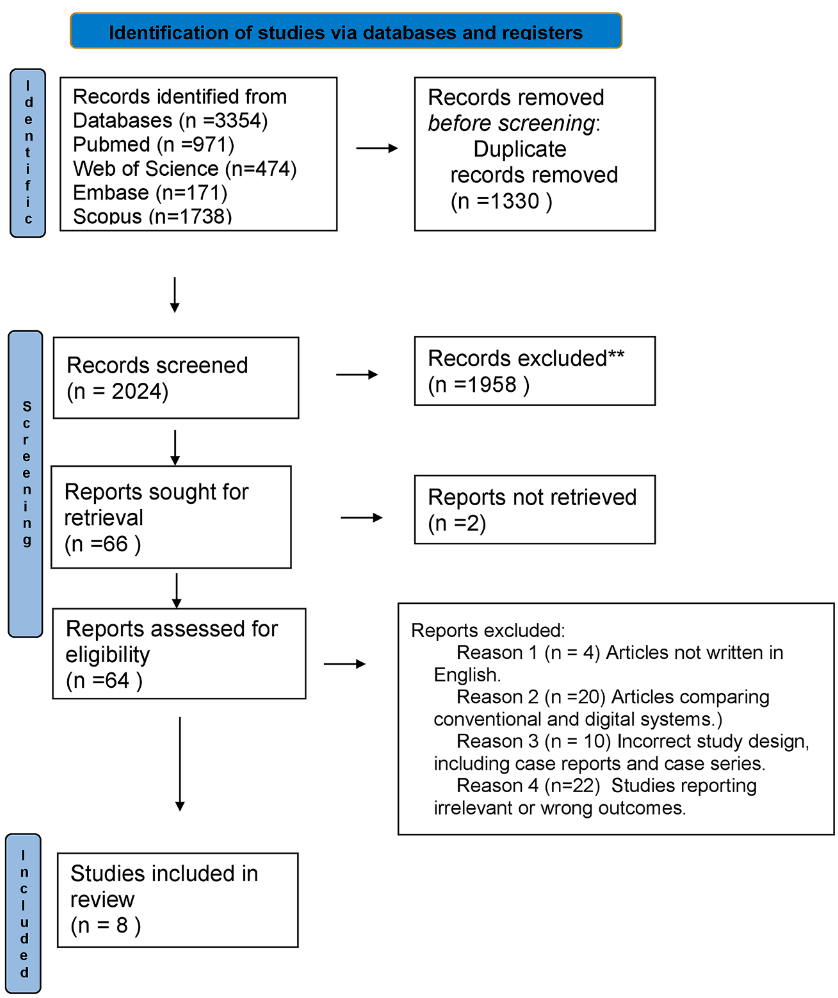

Fig. 1. Search strategy: flow chart of the screening process.

bias as low, medium, or high [25].

3. Results

The PRISMA study selection flowchart is shown in Fig. 1. A total of 3354 articles were retrieved, and after removing duplicates ($n = 1330$), 2024 articles were screened for title and abstract. From these, 1960 were excluded, leaving 64 for full-text evaluation. Ultimately, eight studies met the eligibility criteria and were included in the review (8, 9, 13, 16, 18, 20, 22, 23) (Table 3). The studies were conducted in South Korea ($n = 2$), Bulgaria ($n = 1$), Turkey ($n = 1$), China ($n = 1$), Taiwan ($n = 1$), Spain ($n = 1$), and Romania ($n = 1$). Two studies were classified as in vitro [18,20], while the others were in vivo [8,9,13,16,22,23].

3.1. Classification of digital technologies

Based on the digital technologies employed, the included studies were classified into the following categories.

- DOAS: Used in six studies [13,16,18,20].
- IOS combined with DOAS: Employed in four studies [8,9,22,23].

No studies included Desktop Scanners (DS) technology. The digital technologies used were further categorized based on the specific devices employed.

- DOAS: T-Scan (Tekscan Inc., South Boston, MA, USA) versions (III, Novus, v8.0) [8,9,13,18,20,21], OccluSense (Dr. Jean Bausch GmbH & Co KG) [20], Dental Prescale II (DP2 GC Corp., Tokyo, Japan) [16], Accura (Dmetec Co., Bucheon, Korea). (13, 18).
- IOS: CEREC Omnicam ™ (version 4.2, Sirona Dental Systems, NY, USA) [9,21], Trios Standard Pod (3Shape, Copenhagen, Denmark, 2014) [22], Medit I600 (Medit link version 3.1.4, Seoul, Republic of Korea) [8].

Additionally, some studies combined IOS and DOAS technologies, such as T-Scan Novus paired with the Trios Standard Pod [22] and T-Scan III combined with the Medit I600 IOS [8].

Regarding the details of digital scanning in this review, studies used different digital technologies to analyze occlusal contacts and forces with key devices demonstrating specific capabilities and applications.

3.2. T-Scan systems

The T-Scan systems, T-Scan systems (III, Novus, v8.0) were commonly employed to assess the bite forces, occlusal contacts, and pressure distribution [8,9,13,18,20,21]. These devices utilized calibrated pressure-sensitive sensors with sensitivity adjustments for optimizing measurement accuracy. Their high frame rates supported detailed real-time analysis of dynamic occlusal events, such as maximum intercuspation and lateral movements.

3.3. IOS

Several studies [8,9,21,22] used IOS, including CEREC Omnicam, Medit I600 and Trios Standard Pod (3Shape) for generating three-dimensional (3D) digital models of the maxillary and mandibular arches. These devices utilized various technologies, including intraoral scanning (IOS) based on structured light and extraoral stereophotogrammetry, for generating standard tessellation language (STL) files capturing detailed occlusal, buccal, and lingual surfaces in the maximum intercuspal position. Integration of these STL files with T-Scan data permitted the evaluation of occlusal contact points and force distributions, combining quantitative measurements with visual illustration.

3.4. Pressure-sensitive films

The DP2 system was used in the study [16] to complement digital technologies by offering a visual and quantitative assessment of the occlusal forces. Multi-layered films embedded with microcapsules released a dye under pressure, producing a color-coded map of contact intensity. Subsequent analysis with a color image scanner quantified the bite forces, occlusal contact areas, and pressure distributions, providing a clear perspective on the occlusal load.

3.5. OccluSense systems

The OccluSense system, equipped with piezoelectric film sensors, was used to assess the occlusal contact forces and visualize their distribution using color-coded maps. High-intensity contacts were represented in red, while blue indicated low-intensity contacts, presenting a detailed assessment of occlusal balance and potential points of overload [20].

3.6. Combination methods

Several studies [8,9,21,22] combined IOS (e.g., Trios, CEREC Omnicam) with T-Scan systems to enhance the assessment of occlusal force assessment. This integration allowed advanced visualization of force distributions superimposed on 3D dental models, delivering detailed insights into occlusal contact patterns, balance and force distribution. Such approaches demonstrated the potential of combined methods in improving the accuracy and reliability of the occlusal analysis.

3.7. Reliability and consistency of the measurements

Only one study explicitly evaluated reliability using intra-class correlation coefficients (ICC) and inter-examiner correlation coefficients (IECC), reporting high reliability for the Accura and T-Scan Novus systems [18]. Most studies did not rigorously assess measurement reliability, indicating a gap warranting further standardization and methodological improvement. A study-level quantitative overview of the reported indices (e.g., ICC/IECC, sensitivity/specificity) is compiled in Supplementary Table S2 to facilitate rapid cross-study appraisal.

3.8. Comparative performance of DOAS

Across the included studies, T-Scan consistently provided dynamic force-timing information and higher sensitivity to low-intensity contacts, particularly in the posterior regions, whereas IOS-based systems (e.g., CEREC Omnicam) were more accurate for static localization—especially in the anterior region; however, they lacked temporal metrics [8,9,21,22]. DP2 offered absolute bite-force quantification without contact-sequence analysis, and OccluSense exhibited lower precision than T-Scan, with partial improvement when centering support was used [16,20]. Accura demonstrated acceptable diagnostic performance and high reliability against a universal testing reference, although with more limited analytic depth than T-Scan [13,18]. Integration of IOS + *T*-Scan data enhanced the assessment by superimposing dynamic forces on three-dimensional models, improving the interpretation of contact patterns and right–left balance [8,9,22]. One study also noted larger contact areas in men than women [9]. A cross-system summary of primary outputs, strengths, limitations, and key references is shown in Table 4.

3.9. Risk of bias

A structured assessment was conducted to evaluate the risk of bias separately for in vitro and cross-sectional studies. The primary factors contributing to bias included confounding variables, inadequate

Table 3
Study characteristics.

First Author (Year)	Study Design	Number of Patients / Samples	Digital Technology	Technology Details	Sensor Thickness
Jeong (2020) [13]	In vivo (cross-sectional)	24 patients	DOAS (T-Scan v8.0, Accura)	T-Scan v8.0; Accura	T-Scan: 100 µm, Accura: 160 µm
Bostancıoğlu (2021) [9]	In vivo (cross-sectional)	40 patients	IOS and DOAS (CEREC Omnicam, T-Scan III)	CEREC Omnicam v4.2; T-Scan III v9.1	T-Scan: 100 µm, CEREC: NA
Cao (2023) [21]	In vivo (cross-sectional)	20 patients	IOS and DOAS (CEREC Omnicam, T-Scan III)	CEREC Omnicam; T-Scan III v8.0	T-Scan: 100 µm, CEREC: 100 µm
Huang (2022) [16]	In vivo (cross-sectional)	80 patients	DOAS (DP2, T-Scan III)	Dental Prescale II (DP2); T-Scan III v8.0	DP2: 150 µm, T-Scan: 60 µm
Lee (2022) [18]	In vitro	1 phantom	DOAS (Accura, T-Scan Novus)	Accura; T-Scan Novus	NR
Jauregi (2023) [20]	In vitro	10 phantoms	DOAS (T-Scan Novus, OccluSense)	T-Scan Novus; OccluSense	NR
Shopova (2023) [22]	In vivo (cross-sectional)	32 patients	IOS and DOAS (3Shape Trios, T-Scan Novus)	Trios; T-Scan Novus	NR
Popa (2024) [8]	In vivo (exploratory cross-sectional)	20 patients	IOS and DOAS (Medit I600, T-Scan III)	Medit I600; T-Scan III	T-Scan: 60 µm, Medit I600: NA

First Author (Year)	Device Calibration	Scan Protocols	Bite Position	Superimposition & Software	Anatomical Reference
Jeong (2020) [13]	NR	T-Scan & Accura; central incisor width measured; sensor aligned with occlusal plane.	MIP	ImageJ; occlusal casts, silicone records, T-Scan & Accura overlaid using arch dimensions.	Occlusal surfaces of maxillary teeth divided into regions.
Bostancıoğlu (2021) [9]	NR	T-Scan and CEREC; identical head positioning; buccal and intraoral images acquired.	MIP	Adobe Photoshop CS6; image processing, histogram analysis for contact comparison.	Occlusal surfaces of all maxillary teeth.
Cao (2023) [21]	NR	CEREC for 3D models; T-Scan arch model generated from incisor width.	MIP	Photoshop CS6; articulating paper images used for alignment.	Occlusal surfaces of all maxillary teeth.
Huang (2022) [16]	NR	T-Scan & DP2; T-Scan used first, then DP2 film after rest period.	MIP	NA	Occlusal surfaces of all maxillary teeth.
Lee (2022) [18]	NR	Accura and T-Scan Novus; sensors tested 3×/device/day by 2 examiners.	MIP	System software; aligned with universal testing machine; data analyzed via X-Y coordinates.	Occlusal contact points.
Jauregi (2023) [20]	NR	T-Scan & OccluSense; mounted on articulator; piezoelectric sensor used.	MIP	T-Scan & OccluSense software; superimposed contact maps; force balance across midline.	Balance of occlusal contact forces relative to midline.
Shopova (2023) [22]	NR	Trios and T-Scan; STL files integrated; incisor size and missing teeth noted.	MIP	NA	Dental arch STL data integrated with occlusal scans.
Popa (2024) [8]	Yes	T-Scan III (sensitivity adjusted); Medit I600 (zigzag scan); occlusal maps recorded.	MIP	Photoshop CS6; JPEG conversion, anatomical landmarks used for manual alignment.	NA

First Author (Year)	Measurement Methodology	ICC Reported	IECC Reported	ICC Value	IECC Value	Key Outcomes	Main Conclusions
Jeong (2020) [13]	Silicone interocclusal records; alignment by arch length and width	NR	No	NA	NA	Accura vs. T-Scan: Accuracy: 75.8 %, Sensitivity: 82.1 %, Specificity: 60.1 %, PPV: 82.9 %, NPV: 60.1 %	Accura could serve as an alternative to T-Scan for identifying occlusal contacts.
Bostancıoğlu (2021) [9]	Color-coded contact areas using T-Scan and CEREC	NR	No	NA	NA	T-Scan identified more contact areas; gender	T-Scan was more sensitive than CEREC Omnicam,

(continued on next page)

Table 3 (*continued*)

First Author (Year)	Measurement Methodology	ICC Reported	IECC Reported	ICC Value	IECC Value	Key Outcomes	Main Conclusions
	Omnicam; tolerance range set at 8 %					analysis showed men had larger contact areas.	particularly in light contact detection.
Cao (2023) [21]	Occlusal contact area via pixel counts; overlap ratios calculated	NR	No	NA	NA	CEREC Omnicam more accurate in anterior region; T-Scan more reproducible posteriorly.	CEREC Omnicam excelled in anterior accuracy; T-Scan superior in posterior reproducibility.
Huang (2022) [16]	T-Scan (relative force); DP2 (absolute force in N); comparative correlation	NR	No	NA	NA	Significant correlation between systems; T-Scan offers dynamics, DP2 provides absolutes.	Combining T-Scan and DP2 may improve bite force evaluation in clinical settings.
Lee (2022) [18]	Testing via universal testing machine; ICC and IECC assessed across days and raters	Yes	Yes	T-Scan: 0.963, Accura: 0.952; Repeatability: T-Scan: 0.911, Accura: 0.938	T-Scan: 0.991/0.931, Accura: 0.919/0.974	Both systems showed high reliability and repeatability.	T-Scan and Accura demonstrated excellent performance; T-Scan slightly superior.
Jauregi (2023) [20]	GRR test on articulator; AIAG standards used for classification	NR	No	NA	NA	T-Scan: Total variation 0.8 % (classified as high precision); OccluSense: 12 % (low precision)	T-Scan showed superior repeatability and reproducibility over OccluSense.
Shopova (2023) [22]	T-Scan with or without pre-imported scans; anterior tooth width used if scans absent	NR	No	NA	NA	Trios limited to proximity-based contacts; T-Scan detected contact strength, timing.	T-Scan is preferable for complex cases and dynamic occlusal analysis.
Popa (2024) [8]	T-Scan III (dynamic analysis) vs. Medit I600 (color-coded static map)	NR	No	NA	NA	Systems provided different outputs; Medit suitable for morphology, T-Scan for force.	T-Scan remains gold standard; further studies needed for Medit I600 interpretation.

Table 4
Comparative overview of digital occlusal analysis systems.

System	Primary Output	Strengths	Limitations	References
T-Scan (III, Novus, v8.0)	Dynamic occlusal force, timing, sequence	High sensitivity; real-time analysis; reproducible results	Relative force only; sensor thickness may affect anterior accuracy	[8,9,13,18,20,21]
Accura	Occlusal contact detection, relative force	Cost-effective; acceptable sensitivity/specificity	Limited dynamic data; relative rather than absolute force	[13,18]
CEREC Omnicam	Static occlusal contacts (3D STL)	High accuracy in anterior region; integrates with CAD/CAM	Poor reproducibility in posterior contacts; no force/timing data	[9,21]
Trios (3Shape)	Static occlusal contacts (3D STL)	Quick chairside scans; integration with T-Scan enhances analysis	Limited to static visualization; cannot assess force dynamics	[22]
Medit I600	Static occlusal morphology (3D STL)	Easy use; detailed contact morphology	Lacks dynamic occlusal data; limited reliability indices reported	[8]
DP2 (Dental Prescale II)	Absolute bite force (N), contact areas	Provides quantitative force values; strong correlation with T-Scan	No temporal/sequential analysis; single-use films	[16]
OccluSense	Force distribution, color-coded maps	Portable; intuitive visualization	Lower precision than T-Scan; reproducibility issues, improved but not resolved with centering support	[20]

Table 5
Risk of bias assessment according to the Joanna Briggs Institute Checklist for Prevalence Studies.

Studies	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	TOTAL (%)	Risk of Bias
Jeong (2020) [13]	U	YES	U	YES	YES	YES	YES	YES	U	66.66 %	Moderate
Bostancıoğlu (2021) [9]	YES	YES	YES	YES	YES	YES	YES	YES	U	88.88 %	Low
Cao (2023) [21]	YES	YES	U	YES	U	YES	YES	YES	U	66.66 %	Moderate
Huang (2022) [16]	YES	YES	NO	YES	YES	YES	YES	YES	U	77.77 %	Low
Shopova (2023) [22]	YES	NO	NO	YES	U	YES	YES	YES	U	55.55 %	Moderate
Popa (2024) [8]	U	NO	NO	YES	YES	U	YES	YES	U	44.44 %	High

Legend:.
Q1: Was the sample frame appropriate to address the target population?.
Q2: Were study participants sampled in an appropriate way?.
Q3: Was the sample size adequate?.
Q4: Were the study subjects and the setting described in detail?.
Q5: Was the data analysis conducted with sufficient coverage of the identified sample?.
Q6: Were valid methods used for the identification of the condition?.
Q7: Was the condition measured in a standard, reliable way for all participants?.
Q8: Was there appropriate statistical analysis?.
Q9: Was the response rate adequate, and if not, was the low response rate managed appropriately?.
TOTAL= ΣYes / 9 (Total of items) x 100 %.
Rating and cut-offs: The risk of bias was classified as low when the study reached ≥70 % score for "yes," moderate when the study reached 50 % - 69 % score for "yes," and high when the study reached ≤49 % score for "yes.

blinding, and insufficient reporting of randomization.

For cross-sectional studies, the JBI Checklist for Prevalence Studies was used to assess methodological quality. Two studies [9,16] demonstrated a low risk of bias, as they provided comprehensive methodology descriptions and employed appropriate statistical analyses. Three studies [13,21,22] were classified as having a moderate risk of bias because of limited details regarding sample size calculations and the absence of blinding procedures. One study (8) was categorized as high risk as it lacked a justified sample size and did not incorporate randomization, raising concerns about the reliability of its findings. The detailed justification for these classifications is presented in Table 5, Supplementary Table S3.

For in vitro studies, the QUINN tool was used to evaluate bias, revealing moderate limitations, particularly in randomization and blinding. One of the studies [18] exhibited strong overall reliability but had methodological inconsistencies, leading to a moderate risk of bias classification. Similarly, another study [20] faced limitations concerning sample representativeness and statistical validation, contributing to a moderate-risk assessment. The specific reasons for these ratings are outlined in Table 6, Supplementary Table S4.

4. Discussion

The assessment of occlusal relationships plays a critical role in maintaining functional balance within the masticatory system and optimizing clinical outcomes across dental disciplines. Traditional methods such as articulating paper, Shimstock foil, and occlusal sprays, are widely used but are limited by their qualitative nature and inability to measure contact force or timing dynamically. Research has consistently highlighted their limitations, including poor reliability, susceptibility to external factors such as saliva, and the production of false markings [26–29].

Occlusal contact analysis, which includes the evaluation of contact positions, number of occlusal contacts, and occlusal force applied by each individual tooth, plays an important role in dentistry, especially within the realms of prosthodontics and orthodontics, since occlusal adjustment is aimed at maintaining the stability of occlusal contacts and forces related to the teeth or prosthetic appliances. However, a practical and reliable approach capable of simultaneously acquiring the aforementioned parameters associated with occlusal contact is lacking [30].

Recent advancements in digital technologies have addressed many of these limitations, providing clinicians with more precise and reproducible data for occlusal analysis [31]. This systematic review aimed to evaluate the accuracy, reliability, and clinical efficacy of various digital occlusal analysis technologies in dentate arches. The findings revealed significant differences in the performance, application, and clinical outcomes across these systems. These variations highlight the importance of selecting the appropriate technology based on specific clinical needs and patient scenarios, ultimately influencing treatment planning and outcomes.

Table 6
Risk of bias assessment using the OUIN Tool.

References	Lee (2022) [18]	Jauregi (2023) [20]
Clearly stated aims/objectives	2	2
Detailed explanation of sample size calculation	0	0
Detailed explanation of sampling technique	0	0
Details of comparison group	2	1
Detailed explanation of methodology	1	2
Operator details	2	1
Randomization	0	0
Method of measurement of outcome	1	2
Blinding	0	0
Statistical analysis	2	1
Presentation of results	1	2
Score	11	11
%	50	50
Risk of bias	medium	medium

The included studies demonstrated that T-Scan systems, including versions III and Novus, consistently provided detailed insights into occlusal force distribution, contact sequences, and dynamic occlusal patterns. These systems offered high repeatability and reliability, with sensitivity adjustments and calibration improving accuracy. For instance, Lee et al. [18] reported excellent intra- and inter-examiner reliability for T-Scan Novus and Accura systems, with T-Scan Novus slightly outperforming Accura in repeatability metrics.

In contrast, CEREC Omnicam demonstrated higher accuracy in detecting occlusal contact points in the anterior region but lacked reproducibility in the premolar and molar areas compared to that by T-Scan. Cao et al. [21] found that the overlap ratio for CEREC Omnicam was superior to that via T-Scan in the anterior region, while T-Scan excelled in the dynamic analysis of posterior contacts.

Accura was identified as a cost-effective alternative, offering acceptable sensitivity and specificity in occlusal contact detection. However, its reliance on relative force measurements limits its use for more complex occlusal assessments when compared to T-Scan.

Dynamic occlusal analysis systems, particularly T-Scan Novus, excelled in evaluating occlusal timing, force distribution, and sequential contact patterns during functional movements. On the other hand, static occlusal analysis systems, such as DP2 and OccluSense, focused on force magnitude and contact areas but lacked the ability to analyze temporal contact sequences.

The combination of T-Scan with IOS (e.g., Trios, Medit I600) enhanced the accuracy and comprehensiveness of occlusal analysis. Studies by Shopova et al. [22] and Popa et al. [8] highlighted the advantages of integrating dynamic and static analysis systems for a holistic assessment of occlusal parameters.

The reviewed studies underscored the clinical relevance of DOAS in various dental applications, including prosthodontics, orthodontics, and temporomandibular disorder (TMD) management. Jauregi et al. [20] reported that T-Scan systems provide superior precision and reproducibility than does OccluSense, particularly in diagnosing occlusal imbalances.

However, limitations were identified across systems. For instance, IOS, such as Medit I600 provided reliable static occlusal data but lacked the capability to measure dynamic force distribution accurately. Similarly, pressure-sensitive systems such as DP2 excelled in quantifying absolute force values but failed to capture occlusal timing sequences.

This review included only studies published in English, which may have limited the inclusion of potentially relevant studies reported in other languages. This decision was made to ensure methodological consistency and reliable data extraction, considering the lack of access to professional translation services. Future systematic reviews could benefit from a multilingual search strategy to broaden the evidence base.

Additionally, grey literature was not included in this review. Although resources such as OpenGrey may contribute to reducing publication bias, we focused on peer-reviewed publications to ensure methodological transparency, standardized reporting, and traceability. The use of four comprehensive databases (PubMed, Scopus, Web of Science, and Embase) helped minimize the risk of missing relevant studies. This limitation has now been explicitly noted, and future reviews are encouraged to incorporate grey literature to enhance comprehensiveness.

Furthermore, the risk of bias ratings obtained from the JBI and QUIN Tool assessments were carefully considered during evidence synthesis. While all eligible studies were included in the qualitative analysis, findings from studies with a high risk of bias were interpreted with caution and assigned proportionally less weightage in arriving at the main conclusions. Discrepant findings from these studies were presented as preliminary observations or hypothesis-generating rather than being shown to be definitive. As no meta-analysis was conducted, the influence of bias was addressed narratively to ensure transparency and minimize distortion of the review's overall findings.

Methodological shortcomings—insufficient examiner calibration, lack of a priori sample-size justification, inconsistent reporting of reliability (ICC/IECC), and heterogeneous outcomes—limited the comparability and generalizability across studies [8,13,18,21,22]. As summarized in Supplementary Table S2, reporting of reliability and diagnostic performance metrics (e.g., ICC/IECC model specification with 95 % confidence intervals (CIs); the sensitivity/specificity thresholds) was heterogeneous, which precluded performing a meta-analysis and underscores the need for standardized calibration protocols and minimum reporting sets.

To improve the cross-study comparability in digital occlusal research, future studies are advised to:

(i) report calibration procedures (e.g., sensor pre-conditioning, sensitivity wizard, frequency and acceptance limits);
(ii) specify sensor characteristics such as nominal thickness, sampling rate, usable force range, and replacement schedule;
(iii) predefine scan protocols (e.g., MIP, protrusive, laterotrusive movements, number of trials, and rest intervals);
(iv) adopt core outcome sets including balance of occlusal force, first-contact time, sequence to MIP, relative and absolute force with units, contact area/overlap ratio with pixel-to-mm calibration, reproducibility indices);
(v) report reliability metrics using intraclass correlation coefficients (ICC;two-way random-effects, absolute-agreement, 95 % CIs) and inter-examiner agreement.
(vi) detail data processing proceduresincluding filters, frame rates, exclusion criteria, handling of artefacts or missing data);
(vii) ensure blinding of assessors where feasible.
(viii) provide a priori sample-size calculations and ensure transparent statistical reporting.

Adopting these elements will enhance reproducibility and enable quantitative synthesis across platforms.

Beyond methodological standardization, research priorities include adequately powered, multi-center clinical trials that report both accuracy/reliability and efficiency outcomes; integration of AI-assisted analytics; hardware innovations (thinner, less intrusive sensors with higher sampling rates and improved linearity); and native interoperability between IOS/DOAS and CAD/CAM platforms to streamline chairside workflows and enable real-time occlusal refinement. To support future meta-analyses, studies should pre-specify ICC models (two-way random-effects, absolute-agreement) with 95 % confidence intervals, report IECC where applicable, and provide diagnostic accuracy measures (sensitivity/specificity) with clearly defined thresholds (see Supplementary Table S2).

5. Conclusions

This review highlights the strengths and limitations of commonly used DOAS. T-Scan systems enable highly reproducible results for dynamic occlusal analysis, while CEREC Omnicam and Accura offer viable static assessments as alternatives. Combining static and dynamic systems may offer a more comprehensive clinical evaluation. However, the interpretation of results was influenced by the methodological quality of the included studies. Findings from studies with a high risk of bias were interpreted with caution and given less weight in the overall synthesis. Therefore, while current evidence supports the utility of digital occlusal systems in various clinical contexts, future studies with rigorous designs and standardized methodologies are warranted to validate these outcomes and strengthen the evidence base.

CRediT authorship contribution statement

Gökçen Dinçer: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Conceptualization. **Camila Nogueira Chamma-Wedemann:** Investigation, Formal analysis, Data curation. **Isabella Neme Ribeiro dos Reis:** Writing – review & editing, Validation, Formal analysis, Data curation. **Emily Vivianne Freitas da Silva:** Writing – review & editing, Validation, Methodology. **Gülce Çakmak:** Writing – review & editing, Validation, Formal analysis. **Burak Yılmaz:** Writing – review & editing, Validation, Supervision. **Newton Sesma:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jdent.2025.106114.

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